KNECK OFFICIAL

Biology — Paper 1

231/1 • ORIGINAL REVISION SET 01 • 2 hours • 80 marks

Name: _____ Class: _____ Date: _____

Independent KCSE-style practice. Not issued, endorsed or predicted by KNEC. These are original questions, not leaked or past official papers. Component numbering is checked against the KNEC 2026 timetable. A teacher should review these practice materials before formal classroom assessment.

CANDIDATE INSTRUCTIONS

Answer all questions. Use biological terms precisely and show calculations. All questions carry 4 marks.

Use a separate answer booklet. Show working where appropriate. The marking guidance begins after the END OF QUESTIONS page; do not consult it until you finish. Practical and project components require the stated equipment and teacher supervision, not merely a printed worksheet.

SHORT STRUCTURED QUESTIONS

Answer all twenty.

1. State two differences between plant and animal cells and explain one function of a mitochondrion. [4 marks]

2. Define diffusion and osmosis. [4 marks]

3. A cell image is 45 mm long; the actual cell is 0.030 mm. Calculate magnification; state one reason for staining cells. [4 marks]

4. Explain why salivary amylase stops working in the stomach. [4 marks]

5. Give four adaptations of an alveolus for gas exchange. [4 marks]

6. State two functions of blood plasma and two functions of red blood cells/haemoglobin. [4 marks]

7. Explain how stomata close when guard cells lose water. [4 marks]

8. State four factors affecting transpiration and the direction of each effect. [4 marks]

9. Distinguish aerobic from anaerobic respiration in terms of oxygen and energy yield. [4 marks]

10. In peas, T is dominant tall and t dwarf. Cross Tt × Tt: give gametes, genotypes and phenotype ratio. [4 marks]

11. Explain natural selection in a population exposed to a new predator. [4 marks]

12. State four features of insect-pollinated flowers. [4 marks]

13. Describe the roles of the placenta and umbilical cord. [4 marks]

14. Explain four ways the skin helps regulate temperature. [4 marks]

15. State four differences between a sensory and motor neurone. [4 marks]

16. A quadrat of 0.25 m^2 contains 12,8,10,6 and 14 seedlings at five random placements. Estimate density per m^2 and population in a uniform 200 m^2 plot. [4 marks]

17. Explain why decomposers are important in an ecosystem. [4 marks]

18. Name two nitrogen-fixing organisms/groups and distinguish fixation from nitrification. [4 marks]

19. Explain how vaccination produces protection against a pathogen. [4 marks]

20. Give two transmission routes and two prevention measures for cholera. [4 marks]

END OF QUESTIONS

MARKING GUIDANCE

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Suggested answers and assessment guidance, not an official marking scheme. Accept equivalent correct reasoning, valid examples and properly justified alternative methods. Marks for an item must not exceed its printed allocation. For essays use relevant textual evidence and the rubric; no plot summary alone.

SHORT STRUCTURED QUESTIONS

1. (4 marks) Plant cells usually have cellulose wall/large permanent vacuole/chloroplasts, animal cells lack corresponding structures; two paired differences (2). Mitochondrion is site of aerobic respiration/ATP production (2).

2. (4 marks) Diffusion: net movement of particles from higher to lower concentration (2). Osmosis: net movement of water through a selectively permeable membrane from higher to lower water potential (2).

3. (4 marks) M=image/actual (1); 45/.030=1500× (2); increase contrast/make structures visible (1).

4. (4 marks) Stomach acid gives low pH (1); amylase optimum near neutral (1); altered active-site shape/denaturation in strong acidity (1); substrate no longer fits (1).

5. (4 marks) Thin wall; moist lining; large total surface area; dense capillary network/ventilation maintaining diffusion gradient. Any four valid distinct points.

6. (4 marks) Plasma transports nutrients/hormones/CO2/heat/urea (two). RBC haemoglobin transports O2 and buffers/transports some CO2 (two distinct valid functions).

7. (4 marks) Water leaves by osmosis (1); turgor falls (1); guard cells become flaccid (1); pore closes because of differential wall thickness/shape (1).

8. (4 marks) Higher temperature increases; higher humidity decreases; stronger wind usually increases; higher light promotes opening and increases. Accept correctly qualified alternatives.

9. (4 marks) Aerobic requires O2, anaerobic does not (2). Aerobic releases much more energy per glucose; anaerobic only partial breakdown and less ATP (2).

10. (4 marks) Gametes T,t each (1); TT,Tt,Tt,tt (2); 3 tall:1 dwarf (1).

11. (4 marks) Heritable variation (1); some variants better evade predator (1); differential survival/reproduction (1); favourable alleles become more common over generations (1).

12. (4 marks) Large/conspicuous petals; scent; nectar; sticky/spiny pollen; small quantity heavy pollen; sticky stigma inside. Any four.

13. (4 marks) Placenta exchanges O2/nutrients from maternal side and wastes/CO2 to maternal side without normally mixing blood (3). Cord contains vessels connecting fetus and placenta (1).

14. (4 marks) Sweating/evaporation; vasodilation increases heat loss; vasoconstriction reduces heat loss; insulation by subcutaneous fat/hairs with correct context. Four distinct mechanisms.

15. (4 marks) Sensory receptor→CNS, motor CNS→effector; sensory cell body off fibre in ganglion versus motor in CNS; sensory long dendron/short axon versus motor short dendrites/long axon; entry dorsal versus exit ventral spinal root. One per paired difference.

16. (4 marks) Mean=50/5=10 (1); density=10/.25=40/m² (1); population=40×200=8000 (2).

17. (4 marks) Break down dead organic matter (1); recycle mineral nutrients (1); return carbon through respiration (1); make nutrients available to producers/maintain soil fertility (1).

18. (4 marks) Rhizobium and free-living Azotobacter/certain cyanobacteria (2). Fixation converts N2 to biologically available nitrogen; nitrification oxidises ammonium to nitrite then nitrate (2).

19. (4 marks) Introduces harmless antigen (1); stimulates specific immune response/antibodies (1); forms memory cells (1); faster stronger secondary response on exposure (1).

20. (4 marks) Contaminated water and food (2); safe treated water, sanitation, hand hygiene, safe food handling, any two (2).

REVISION CHECK

Mark your work, list three misconceptions, revise those topics and attempt the incorrect questions again without the guide. These topics are revision priorities, not a forecast of the real examination.

Paper/component reference: https://www.knec.ac.ke/wp-content/uploads/2026/02/2026-KCSE-Timetable.pdf